

To calculate the variance of a data set in Excel, you can use the built-in functions **VAR.P** (for population variance) or **VAR.S** (for sample variance). Here's how you can do it along with an example.

Formula for Variance

1. Population Variance (VAR.P)

$$\text{The formula for the population variance } \sigma^2 \text{ is } \sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu_p)^2}{N}$$

Where:

- N = Number of data points (the size of the population)
 - x_i = Each individual data point
 - μ_p = Mean of the population data points
- #### 2. Sample Variance (VAR.S)

$$\text{The formula for the sample variance } s^2 \text{ is } s^2 = \frac{\sum_{i=1}^N (x_i - \mu_s)^2}{N - 1}$$

Where:

- N = Number of data points (the size of the population)
- x_i = Each individual data point
- μ_s = Mean of the sample data points

Step 1: Calculate the Mean (Average)

Step 2: Calculate $x_i - \mu_s$

Step 3: Calculate $(x_i - \mu_s)^2$

Step 4: Sum of $(x_i - \mu_s)^2$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
10	-4.8	23.04
12	-2.8	7.84
14	-0.8	0.64
18	3.2	10.24
20	5.2	27.04

Calculate the Variance:

- **Population Variance:** $\sigma^2 = 68.8/5 = 13.76$
- **Sample Variance:** $s^2 = 68.8/4 = 17.2$

Apply Variance Formula in Excel

- **Population Variance:** =VAR.P(range)
- **Sample Variance:** =VAR.S(range)

